Water Quality Report

Lakeside Buoy 2026 · 2026-10-03 to 2026-10-10

Action Required

Coordinates	44.0450° N, 83.3977° W (Lakeside MI)
Water Body Type	Freshwater (from device registry)
Client / Contract	Not available from device registry
Report Date	2026-10-10
Last Reading	2026-10-10 08:41 UTC (8 minutes before this report)
Prepared By	Clean Earth Rovers
Omitted	threshold notes (at the requester's request)

1. Summary

- Overall status: Action Required.
- Dissolved Oxygen (mg/L) moved outside the site baseline for the 2026-10-03 to 2026-10-10 reporting period.
- No discrete pollution events were identified, though readings moved outside baseline.
- Turbidity (Relative): Clear (relative index mean 0.0, held steady across the period) — operator-authoritative bands over an uncalibrated scale with no operator range; reported as a band, not judged against one. All readings were 0: possibly a missing sensor, not confirmed clear water.
- Recommendation: see Recommendations below for the operational, investigative, and stakeholder follow-up.

2. Parameter Data

Parameter	Site Baseline	Min	Max	Mean	Median	Out of range	Flag
Dissolved Oxygen (mg/L)	5-14	0.00	11.15	8.41	8.60	0%	Exceedance
ORP (mV)	-200-400	180.60	298.50	241.30	246.50	0%	Normal
pH	6.5-9	7.61	8.15	7.89	7.90	0%	Normal
Conductivity (µS/cm)	100-1500	353.00	549.00	448.91	447.00	0%	Normal
Temperature (°F)	40-85	61.26	74.94	67.84	67.91	0%	Normal
Turbidity (Relative)	No range (relative index)	0.00	0.00	0.00	0.00	—	Clear (all zero)

Flag values: Normal, Elevated, Low, or Exceedance relative to the site baseline. Out of range is the share of the period's series buckets whose average sat outside that baseline — approximate, at bucket resolution, and shown because a flag alone cannot separate a single stray reading from a sustained offset. Turbidity is reported as a water-clarity band (Clear / Moderate / Turbid), operator-authoritative as of 2026-09-10, from an uncalibrated relative index derived from a raw sensor voltage. No operator turbidity range exists on any device, so it is never flagged in or out of range; its Min/Max/Mean/Median are shown for period-to-period comparison only. A Flag reading '(all zero)' means every reading in the period was 0, which the backend also reports for a missing or offline sensor, so it is not confirmed clear water; '(off-scale)' means the reading is at or beyond the top of the conversion's scale.

3. Parameter Analysis

Each chart plots the period's series bucket means, oldest to newest. The shaded band is the site baseline where one exists; the dots mark the plotted high and low.

Dissolved Oxygen (mg/L)

Exceedance

Followed a clear diel rhythm; against the 5-14 mg/L operator-set threshold for this device it moved sharply below it to 0.00 mg/L, recorded as an exceedance. Outside baseline in 0% of the period's series buckets.

pH

Normal

Followed a clear diel rhythm, remaining within the 6.5-9 operator-set threshold for this device.

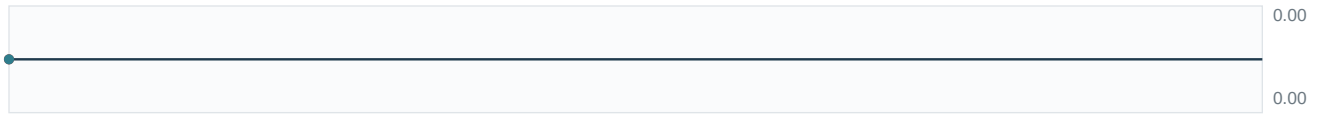
Temperature (°F)

Normal

Followed a clear diel rhythm, remaining within the 40-85 °F operator-set threshold for this device.

Turbidity (Relative)

Clear (all zero)



Water clarity read as Clear for the period (relative index mean 0.0, range 0.0-0.0), held steady across the period. Relative index derived from a raw sensor voltage by a provisional, uncalibrated conversion; no operator turbidity range exists, so this is a clarity band and a direction of change, not a measurement judged in or out of range. Every turbidity reading in this period is 0. The backend reports a missing or offline sensor voltage as 0, the same as clear water, so this may be a missing sensor rather than confirmed clear water.

4. Data Quality

Check	Result	Notes
Data completeness	100.0%	169 of 169 readings returned for the period were usable for at least one parameter. 0 readings were left out because no requested parameter was usable. Values from likely failed sensors were left out. Values with the probe's own fault flag were left out. Values outside plausibility bounds may be sensor faults and were left out. Share of returned readings, not coverage against an expected sampling cadence.
Calibration status	Review	Turbidity has a sustained zero-variance run at exactly 0 or 1005, indicating a likely failed sensor. Affected readings were excluded; inspect the sensor. Readings outside plausibility bounds without a fault flag (Turbidity (Relative): 2) may be sensor faults and were left out. Inspect and recalibrate the affected probes.
Drift indicators	Not assessed	Not assessed — this pipeline has no drift detector. Requires comparison against a calibration record, which is not available from the device API.
Biofouling indicators	Not assessed	Not assessed — this pipeline has no biofouling detector. Requires service history or a fouling-sensitive baseline, neither of which is available here.

Sensor agreement	Not assessed	Not assessed — cross-sensor agreement needs a second co-located pod or a grab-sample result to compare against.

5. Recommendations

Operational: Inspect the sensors for fouling or damage and check that the pod is sitting properly in the water, so an instrument problem can be ruled out. Clean the sensors only if you can do so safely and within your site's safety rules; if a sewage, bloom or chemical release is suspected, avoid touching the water. If the battery is low or a sensor looks damaged, contact Clean Earth Rovers support. Don't let this delay other steps if people, pets or fish may be at risk. Contact: sales@cleanearthrovers.com, or your usual contact at Clean Earth Rovers if you already work with one

Investigative: Ask a qualified person to confirm the flagged readings with an independent measurement, such as a calibrated handheld meter or a laboratory sample, before drawing conclusions. Avoid contact with water that may be contaminated. Keep watching the readings over the following days. A change that lasts across several readings or recovers gradually is more likely real. A single isolated point is more often a sensor glitch, but a brief spike can also be real, so note when it happened.

Stakeholder: Share this report with the people responsible for the site, such as your site manager or monitoring team.